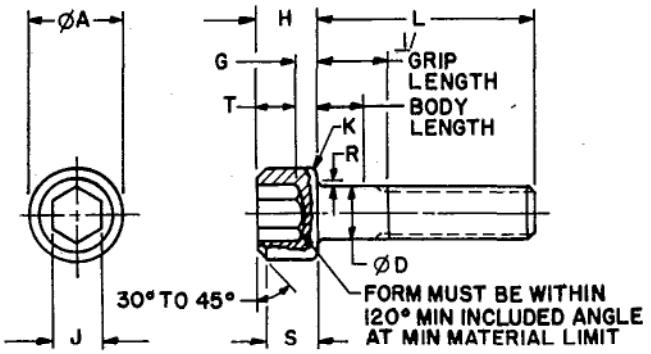
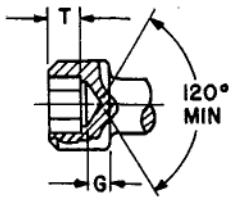




ACCEPTABLE SOCKET DESIGN

DASH NO	1/ GRIP LENGTH MAX
03	.375
08	.500
09	1.000
10	1.500
19	.500
20	1.000
21	1.500
26	.750
27	.750
28	2.250

TABLE I. DIMENSIONS AND PART NUMBERS.

NOMINAL SIZE			.138	.190	.250	.3125	.375	.500	.750
THREADS PER INCH UNC - 3A			32	24	20	18	16	13	10
Ø D - BODY DIAMETER		MAX	.1380	.1900	.2500	.3125	.3750	.5000	.7500
		MIN	.1329	.1840	.2435	.3053	.3678	.4919	.7406
Ø A - HEAD DIAMETER		MAX	.2260	.3125	.3750	.4688	.5625	.7500	1.1250
		MIN	.2180	.3030	.3650	.4570	.5500	.7350	1.1070
H - HEAD HEIGHT		MAX	.1380	.1900	.2500	.3125	.3750	.5000	.7500
		MIN	.1340	.1850	.2440	.3060	.3680	.4920	.7400
S - HEAD SIDE HEIGHT		MIN	.1240	.1710	.2250	.2810	.3370	.4500	.6750
J - SOCKET WIDTH ACROSS FLATS		MAX	.1111	.1587	.1900	.2530	.3160	.3790	.6310
		MIN	.1094	.1562	.1875	.2500	.3125	.3750	.6250
T - KEY ENGAGEMENT		MIN	.0640	.0900	.1200	.1510	.1820	.2450	.3700
G - WALL THICKNESS		MIN	.0470	.0650	.0950	.1190	.1430	.1900	.2850
K - CHAMFER OR RADIUS		MAX	.0050	.0050	.0080	.0080	.0080	.0100	.0100
R - FILLET EXTENSION		MAX	.0100	.0140	.0140	.0170	.0200	.0260	.0390
		MIN	.0060	.0090	.0090	.0120	.0150	.0200	.0300
MINIMUM TENSILE LOAD - LBS			1,450	2,800	5,100	8,400	12,400	22,700	53,400
L - LENGTH	TOLERANCE		DASH NO	DASH NO	DASH NO	DASH NO	DASH NO	DASH NO	DASH NO
.375 .500 .750 1.000	+.000 -.031		01	02	04 05 06	11 12	15 16	22	
1.250 1.500 1.750 2.000 2.500	+.000 -.060			03 1/	07 08 1/ 09 1/ 10 1/	13	17 18 19 1/ 20 1/	23 24 25 26 1/	29
2.750 3.000 4.000	+.000 -.080					14	21 1/	27 1/ 28 1/	

1/ GRIP LENGTH MAX

USER ACTIVITIES:
ARMY - ME, MI
NAVY - MC

REVIEWER ACTIVITIES:
ARMY - AV, EA
NAVY - OS
AIR FORCE - 82
DLA - IS
NSA - NS

APPROVED 1 AUG 84
REVISED

This military standard is approved for use by all Departments and Agencies of the Department of Defense. Selection for all new engineering and design applications and for repetitive use shall be made from this document when applicable.

P.A.	INTERNATIONAL INTEREST	TITLE	MILITARY STANDARD
AR		SCREW, CAP, SOCKET HEAD, ALLOY STEEL, UNC-3A	MS51484
Other Cust SH 99			
PROCUREMENT SPECIFICATION: FF-S-86	SUPERSEDES:		PAGE 1 OF 2

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T18:02Z
Check the source to verify that this is the current version before use.

REQUIREMENTS:

1. MATERIAL. ALLOY STEEL IN ACCORDANCE WITH PROCUREMENT SPECIFICATION HAVING A MINIMUM TENSILE STRENGTH OF 160,000 PSI.

2. FINISH. ZINC COATING IN ACCORDANCE WITH ASTM B633-78, Fe/Zn5 (5 µm), TYPE II.

3. THREADS. THE THREADS SHALL BE IN ACCORDANCE WITH FED-STD-H28/2. ACCEPTABILITY OF SCREW THREAD SHALL BE IN ACCORDANCE WITH FED-STD-H28/20, SYSTEM 21.

4. THREAD LENGTH. SCREWS, DASH NUMBERS 01, 02, 04, 05, 06, 07, 11, 12, 15, 16, 17, 18, 22, 23, 24 AND 25 SHALL HAVE COMPLETE (FULL FORM) THREADS EXTENDING WITHIN TWO THREADS OF THE HEAD AS MEASURED WITH A THREAD RING GAGE. THREADS SHALL NOT EXTEND TO THE R RADIUS NOR SHALL THREADS EXTEND TO AND UNDERCUT THE HEAD OF THE SCREW.

5. PART NUMBER. THE BASIC MS PART NUMBER IS FOLLOWED BY A DASH NUMBER FROM TABLE I.

EXAMPLE: MS51484-01 - SCREW, CAP, SOCKET HEAD, ALLOY STEEL, .138-32UNC-3A, THREAD LENGTH .750, ZINC COATED

NOTES:

1. ALL DIMENSIONS ARE IN INCHES.

2. THE HEAD MAY OR MAY NOT BE KNURLED AT THE MANUFACTURER'S OPTION. TYPE OF KNURLING OPTIONAL.

3. REFERENCED DOCUMENTS SHALL BE OF THE ISSUE IN EFFECT ON DATE OF INVITATIONS FOR BIDS OR REQUEST FOR PROPOSAL, EXCEPT THAT REFERENCED ADOPTED INDUSTRY DOCUMENTS SHALL GIVE THE DATE OF THE ISSUE ADOPTED.

4. FOR DESIGN FEATURE PURPOSES, THIS STANDARD TAKES PRECEDENCE OVER PROCUREMENT DOCUMENTS REFERENCED HEREIN.

USER ACTIVITIES:
ARMY - ME, MI
NAVY - MC

REVIEWER ACTIVITIES:
ARMY - AV, EA
NAVY - OS
AIR FORCE - 82
DLA - IS
NSA - NS

APPROVED 1 AUG 84 REVISED

This military standard is approved for use by all Departments and Agencies of the Department of Defense. Selection for all new engineering and design applications and for repetitive use shall be made from this document when applicable.

P.A.	INTERNATIONAL INTEREST	TITLE	MILITARY STANDARD
AR		SCREW,CAP,SOCKET HEAD, ALLOY STEEL,UNC-3A	MS51484
Other Cust SH 99			
PROCUREMENT SPECIFICATION: FF-S-86	SUPERSEDES:		PAGE 2 OF 2

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T18:02Z
Check the source to verify that this is the current version before use.